

Image Analysis Report

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Executive Summary

The valve and flange assembly exhibit severe atmospheric corrosion characterized by significant metal loss on fasteners and thick rust scale. This level of degradation poses a high risk to pressure containment and structural integrity, requiring immediate intervention and assessment.

Key Findings

#	Finding
1	Severe atmospheric corrosion with heavy rust scale covering the entire valve body and flange assembly.
2	Significant metal loss and rounding of flange bolt heads, indicating compromised structural integrity of the joint.
3	Complete failure and delamination of the protective coating system across the visible components.
4	Localized pitting and surface oxidation on the adjacent pipe section.
5	Possible crevice corrosion at the flange-to-flange interface, currently obscured by heavy scale buildup.
6	Proximity to ground level (gravel) likely creates a corrosive micro-environment that accelerates degradation.
7	The internal condition and actual remaining wall thickness cannot be determined from this external visual inspection.

Recommendations

#	Recommendation
1	Perform ultrasonic thickness (UT) measurement on the valve body and pipe within 72 hours to verify remaining wall thickness.
2	Schedule immediate replacement of all flange bolts and nuts using new B7/2H hardware.
3	Abrasive blast the assembly to Sa 2.5 cleanliness to allow for accurate pit depth measurement and structural assessment.
4	Apply a high-build epoxy coating system compliant with ISO 12944 for C5 environments once the surface is prepared.
5	Install concrete sleepers or pipe supports to provide at least 150mm clearance from the ground.
6	Inspect flange seating surfaces for grooving or pitting during the next scheduled maintenance isolation.
7	Replace the valve and flange assembly if UT results indicate metal loss exceeds the design corrosion allowance (typically 12.5% of nominal).